



A high precision intrusion detection system for network security communication based on multi-scale convolutional neural network

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ABSTRACT

The openness of network data makes it vulnerable to hackers, viruses and other attacks, which seriously threatens the privacy and property security of users. In order to improve the accuracy of the intrusion detection for network security communication, based on the traditional intrusion detection system, combining with the deep learning theory and shortcomings, this paper proposed an intrusion detection system for network security communication based on multi-scale convolutional neural network, and conducted the corresponding experiments on public data sets. The experimental results perform that compared to the intrusion detection system based on Adaboost model and Recurrent Neural Network model, the convergence speed of multi-scale convolutional neural network system is faster, the average error detection rate is reduced by 4.02%, and the average accuracy is improved by 4.37%. The results prove that the intrusion detection system based on multi-scale convolution neural network has a high detection accuracy.

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1. Introduction

The world information is gradually integrated, and data sharing brings network security risks, so in order to protect people's information security when they work and communicate on the Internet, some scholars have begun to study the network security protection [1]. Among them, the intrusion detection system is one of the widely used technical types in the market, which can effectively prevent network intrusion, and ensure the safe operation of the system by detecting and issuing early warning before the attack is launched [2].

However, it is undeniable that the types of hacker network intrusion are becoming more and more sophisticated, so that the commonly used intrusion detection system for the network security also began to become slow, and the accuracy is significantly poor. To find a good solution, this paper introduces the deep learning technique to design an intrusion detection system, and proposes a high precision intrusion detection system based on multi-scale convolutional neural network (MSCNN).

There are two main contributions in this paper. Firstly, the network data mining technology based on deep learning is used to extract features from unordered data to improve the accuracy of the intrusion detection. Secondly, this paper uses MSCNN to solve the compatibility problem between intrusion detection system

and neural network, reduces processing parameters and improves the convergence speed.

This work mainly includes 5 sections. In Section 1, the background, motivations and contributions of this work are introduced. In Section 2, the results of some intrusion detection systems and convolutional neural network in recent years are discussed. In Section 3, the design idea of the high-precision intrusion detection system based on MSCNN is described. In Section 4, the intrusion detection system is tested and its effect in practical application is discussed, and also the proposed one is compared to some baseline methods. In Section 5, some findings in this paper are concluded and the future is presented.

2. Related work

The information integration is the inevitable trend of Internet development. Increasingly open data brings convenience and data security issues [3]. Therefore, in order to avoid the illegal intrusion, some scholars pay more and more attention to network security system. In order to understand the network intrusion detection data, Wei, et al. proposed a three-dimensional visualization interaction method of network intrusion detection data [4]. Umarani, et al. proposed a hybrid anomaly detection system for wireless sensor networks, and the system makes intrusion attacks in sensor networks more secure, universal and accurate [5]. Sedjelmaci, et al. designed an efficient and lightweight intrusion detection mechanism for service-oriented vehicular networks [6].

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Wei, et al. proposed an intrusion detection system combining hierarchical spatiotemporal features, which improved the accuracy of feature set describing network traffic [7]. Datta, et al. have proposed an effective scheme, on the premise of not affecting the effectiveness of intrusion detection system, it shortens the activity time of the intrusion detection system [8]. Zhang, et al. proposed a detection model and an improved adaptive evolution strategy of constrained covariance matrix [9].

As we know, deep learning as a very advanced technique is being studied by a lot of scholars, and it can take some good solutions for the intrusion detection. At present, the deep learning technique has been widely applied to a lot of fields. Kruthiventi, et al. proposed a fully CNN model, established a bottom-up visual attention mechanism model through explicit prediction, and automatically learned features in a hierarchical way to predict saliency graph in an end-to-end way [10]. CNN is used to identify abnormal EEG samples, and it is found that the accuracy of neural network is obviously better than some other models [11]. Ma, et al. proposed a traffic image learning method based on CNN, which can predict the traffic speed of large-scale and whole network with high accuracy. In the acceptable execution time, the average accuracy of the algorithm is significantly improved [12]. Peng, et al. introduced a module composed of coding layer and time pyramid pool layer, and proposed a new network structure, allowing any number of frames as network input, avoiding the risk of losing any important frame [13]. Antipov, et al. led the team to improve the gender and age prediction methods in the existing demographic estimation methods based on CNN, and designed an advanced gender recognition and age estimation model [14]. Zhu, et al. proposed a single event recognition algorithm framework for cold electron microscope images based on deep learning, which realized the integration of automatic particle picking, selection and verification [15]. Kawahara, et al. proposed a neural network framework to solve this question [16].

To sum up, a lot of researches have been done on the intrusion detection system and deep learning. However, the traditional deep learning model has some shortcomings, such as big samples and time cost. In order to find a good solution, this paper tried to design a MSCNN model and apply it to the intrusion detection.

3. Design of the high precision intrusion detection system for network security communication

Neural network can be well used to deal with intrusion behavior. The main reason is that the neural network, so as to train the neural network model and judge whether the unknown data set has intrusion behavior. Although convolutional neural network theory based on deep learning is used to process positioning data, the research of this algorithm is not mature [17]. Therefore, a model for network data service is proposed, it is a model to extract local features and data features according to network traffic classification. The specific framework is shown in Fig. 1.

The MSCNN model has multiple convolution layers, in which there are three layers (Fig. 1). Each layer has three channel sizes. The multi-scale convolution layer and two convolution layers are mutually staggered, and then connected to the pooling layer through a multi-scale convolution layer, and finally connected to the softmax classifier through three fully connected layers. In order to ensure the learning speed of the network, BN (batch normalization, BN) algorithm is used in convolution layer, and Relu function is used as the activation function of convolution layer, the parameters of BN algorithm + Relu function are set to $11 * 11 * 64$, $9 * 9 * 128$, $7 * 7 * 256$ according to different convolution layers. The over fitting problem of full connection layer can be solved by dropout strategy. Among them, multi-scale convolution is to classify the features of different sizes by

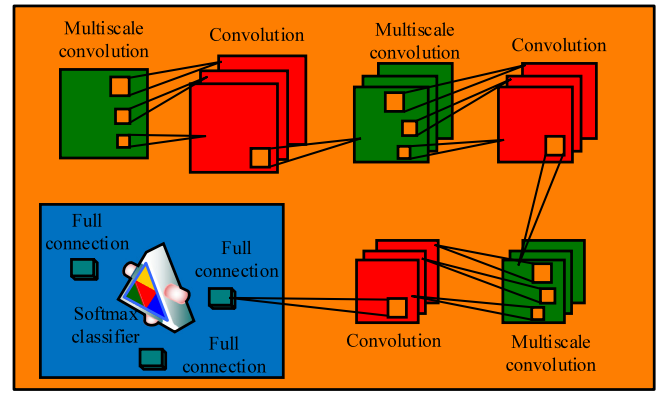


Fig. 1. Multi-scale convolutional neural network model structure.

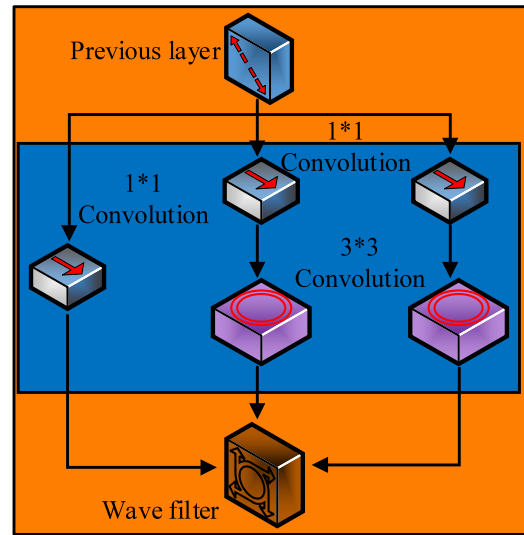


Fig. 2. Multi scale convolution layer structure.

simulating the object features recognized by human vision. The channel sizes used in multi-scale convolution layer are $1 * 1$, and $3 * 3$ respectively, and the structure is shown in Fig. 2.

Firstly, in order to combine the convolution kernel features, set the convolution step size to 1, and then set $\text{pad} = 0, 1, 2$. In addition, the multi-scale convolution layer will reduce the features processed in the previous layer to processing level by $1 * 1$ scale, and then transmits the reduced features to other scale channels (Fig. 2). In this way, the multi parameter problem caused by large-scale convolution kernel convolution can be reduced. Finally, the processing results are sent to the filter for subsequent processing.

The pooling layer in the MSCNN model is also designed to reduce parameters, so as to improve the network training speed, but the pooling layer will lose features [18]. In order to make up for the precision reduction caused by feature loss, the average pooling layer is only added to the last layer of MSCNN model. The advantage of average pooling is that the features in each network data are quite different, there is no uniformity, and the error of classification is small [19]. Overlapping pooling is adopted for pooling of pooling layer. Under certain effect, the convergence speed of pooling layer is improved, and the accuracy is less affected. Therefore, the effect of MSCNN model is the best. The sampling method of pooling layer is shown in Eq. (1).

$$x_j^l = f(\beta_j^l \text{down}(x_j^{l-1}) + b_j^l) \quad (1)$$

In Eq. (1), $down(x_j^{l-1})$ means that the pooling layer samples the output data of the upper layer, β and b represent the feature weight and offset of the output, and $f(*)$ represents the activation function.

The activation function of convolution layer uses Relu function to delete some negative numbers, which increases the non-linearity of the network to some extent, and makes the network structure segment linear. At the same time, the output of some neurons can be 0, which reduces the interdependence between parameters and avoids the over fitting phenomenon of the model [20]. For the full connection layer outside the volume layer, Sigmoid function is used as the activation function to preserve the complete characteristics of the data. The expression of the Relu function is shown in Eq. (2).

$$f(x) = \max(0, x) \quad (2)$$

In Eq. (2), x is the total amount of data, and $\max$ is the maximum value.

In order to avoid the over fitting problem of network, improve the generalization ability of network and speed up the training speed, this study uses Dropout strategy to connect the full connection layer and softmax classifier [21]. The principle of Dropout algorithm is to set the activation value of a neuron to 0 with a certain probability when it propagates forward, so that some neurons stop working, and calculate and connect the data that is not set to 0 [22]. According to the randomly generated probability to connect the neurons, the sparsity of the network appears, the possibility of over fitting is reduced, and the training speed of the network is greatly improved [23]. Under the Dropout strategy, the network is calculated as follows:

$$\begin{cases} r_j^l \sim \text{Bernoulli}(p) \\ \tilde{y}^l = r^l * y^l \\ z_i^{l+1} = w_i^{l+1} \tilde{y}^l + b_i^{l+1} \end{cases} \quad (3)$$

The r_j^l vector in Eq. (3) is generated by Bernoulli function, which will generate a vector of 0 and 1 randomly. y^l is the activation function, and $\tilde{y}^l$ is the activation value after r_j^l vector processing.

In feature extraction and processing of neural network, for the training data with large distribution difference, classification accuracy will be greatly reduced. Because the distribution of each training set is different, if the model needs to adapt each time, it will seriously affect the training speed of the network, so normalization processing is required for each input data [24]. A BN algorithm is proposed, which can be used in volume layer to accelerate the efficiency of network learning and can be used to optimize network training. The BN algorithm is represented as follows:

$$u_b = \frac{1}{m} \sum_{i=1}^m x_i \quad (4)$$

x_i in Eq. (4) represents the data entering the activation function. This step is to calculate the average value of each data. After it is calculated, the variance of each number is calculated

$$\sigma_\beta^2 = \frac{1}{m} \sum_{i=1}^m (x_i - u_\beta)^2 \quad (5)$$

u_β in Eq. (5) represents the mean value, and the variance is σ_β^2 , and then the variance is standardized, as shown in Eq. (6).

$$\hat{x}_i = \frac{x_i - u_\beta}{\sigma_\beta} \quad (6)$$

u_β in Eq. (6) represents the mean value in the sample data, σ_β represents the standard deviation, and then outputs the original value y_i through linear transformation, as shown in Eq. (7).

$$y_i = \gamma \hat{x}_i + \beta = BN_{\gamma, \beta}(x_i) \quad (7)$$

Algorithm: Pseudo code for the MSCNN model building and training process

```

1: Input Trans D MSCNN Model Generate(D)
2: Initial learning rate, Drop
3: Initialize convolution kernel conv_init(N)
4: Initialize pooling layer averagepool_init(size, padding)
5: Initialize full connection layer all_connect_init(size)
6: Dropout Output layer initialization drop (λ)
7: Initialization weights and thresholds(β, b) ∈ (0,1)
8: Initialization function BN
9: MSCNN_Generate(averagepool_init, conv_init, all_connect_init, drop) get the initialization model MSCNN
10: for all d do
11:   if stop criteria not met d then
12:     select small sample subset d ∈ D
13:   end for
14:   input d, to MSCNN, get the current output Y'
15:   for all d do
16:     if d satisfy d ∈ D then
17:       Calculating the loss function C
18:       Update weights β
19:       Update threshold b
20:     end for
21:   output: Update this training CN

```

Fig. 3. Modeling and training pseudo code.

γ and β in Eq. (7) represent training parameters, and BN represents normalization. In the forward propagation, the output value does not change, but changes the two training parameters of γ and β , and each change is recorded for subsequent operation. In the reverse training, the chain derivation is used to calculate the recorded training parameters, and then the learning efficiency of the network is calculated to change the weights.

The mentioned algorithm is used to update the weight and bias of the model. The processing method of gradient descent algorithm is shown in Eq. (8).

$$\begin{cases} \frac{\partial}{\partial w_{ij}^l} J(W, b) = [\frac{1}{m} \sum_{i=1}^m \frac{\partial}{\partial w_{ij}^l} J(W, b; x^i, y^i)] + \gamma W_{ij}^l \\ \frac{\partial}{\partial b_{ij}^l} J(W, b) = \frac{1}{m} \sum_{i=1}^m \frac{\partial}{\partial b_{ij}^l} J(W, b; x^i, y^i) \end{cases} \quad (8)$$

In Eq. (8), w_{ij}^l represents the weights of (i, j) neurons in layer l , b_{ij}^l represents the bias of (i, j) neurons in layer l , and γ represents the training parameters of each layer for parameter regularization. The feedback error is expressed as:

$$\begin{cases} \delta_i^l = (\sum_{j=1}^{s_{l+1}} w_{ij}^l \delta_j^{l+1}) \cdot f'(z_i^l) \\ \delta_i^{n_l} = -(y_i - a_i^{n_l}) \cdot f'(z_i^{n_l}) \end{cases} \quad (9)$$

In Eq. (9), n_l is the last output layer, the residual error of neurons is expressed as δ_i^l , z_i^l is the input value of neurons in layer l , $a_i^{n_l}$ is the activation value of neurons in output layer, s_{l+1} is the number of neuron nodes. The pseudo code of MSCNN model building and training process is shown in Fig. 3.

As shown in Fig. 3, the convolution core, pooling layer and full connection layer need to be initialized first to build the MSCNN model. In the training, the loss function is calculated, and the weights and thresholds are updated continuously. Finally, the updated model after training is obtained.

Intrusion detection system is based on the existing security strategy to detect the network environment before intrusion, in order to early warning to ensure the safe operation of the system [25]. The general working principle of intrusion detection system is shown in Fig. 4.

The intrusion detection system is mainly composed of three parts as shown in Fig. 4. Among them, the system mainly takes

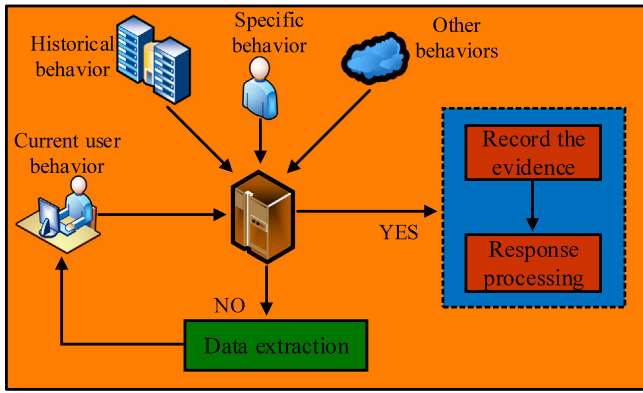


Fig. 4. Working principle of the intrusion detection system.

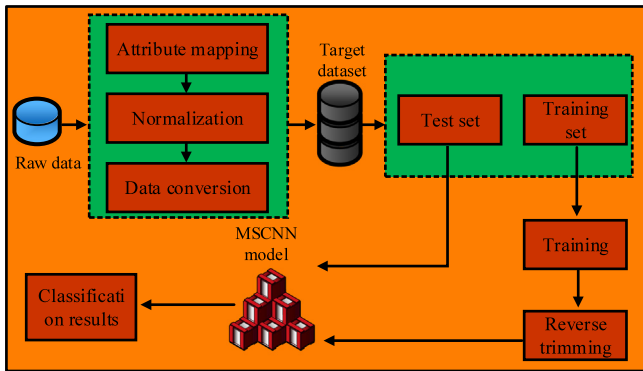


Fig. 5. Learning process of MSCNN.

the historical behavior, specific behavior pattern and other behaviors of monitoring objects as information sources. The information is integrated and transmitted to the analysis engine. The analysis engine detects and analyzes the input information based on the behavior of the system and users to determine whether there is an intrusion behavior. If the result is that there is no intrusion, the normal data is extracted, and the security data is returned to allow the user to operate; if the data has intrusion behavior, the data is sent to the response component to record the evidence and process [26].

However, the current intrusion detection technology is limited by its complexity, resulting in most of the intrusion detection systems still have security risks, such as the lack of defense capability and the accuracy of detection is not optimistic [27]. This paper proposes an intelligent intrusion detection technology combining with MSCNN, and the model learning process is shown in Fig. 5.

As shown in Fig. 5, the original data set is preprocessed. Firstly, the attribute mapping of the data is carried out, the data is digitized, and then normalized to obtain the standardized data set. Finally, the data preprocessing is completed by converting the data into two-dimensional data in accordance with the convolution neural network input format. The integrated target data set is divided into two sub data sets, one is used as the training data set of MSCNN model for model learning, and the weight is adjusted by reverse fine-tuning, the other is the test data of MSCNN model, which is used as the final test analysis to get the final classification results.

The standardization in data preprocessing needs to first digitize the symbolic features, and the operation method of digitization is to use attribute mapping, such as representing the

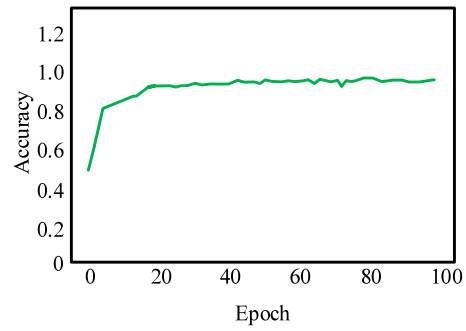


Fig. 6. Model training results.

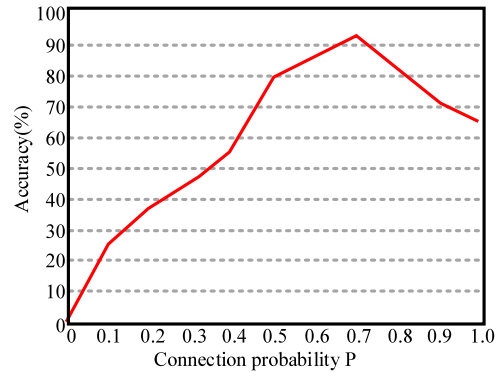


Fig. 7. Comparison results of dropout connection probability selection experiment.

identified protocol type attributes through computer binary coding [28]. The normalization process is to unify the numerical processing results to (0,1) interval distribution, and normalize them through Eq. (10).

$$y = \frac{x - \min}{\max - \min} \quad (10)$$

In Eq. (10), x is the random attribute in the network, $\min$ is the minimum numerical attribute in the network record, and $\max$ is the maximum numerical attribute in the network record. Through the preprocessing of the network data, the data will present the image data format and input to the convolutional neural network.

4. Application effect analysis of the high precision intrusion detection system

The simulation environment is Ubuntu 17.04, the CPU is Intel Core i5-3470, the memory size is 8 GB, the python 3.5 programming language is used for development, and the data reading is pandas. There are 5 million records in this data set. In this experiment, 10% of them are taken as the test set and training set, and then 20% of the selected data set is taken as the training set and 80% as the test set. The training results are shown in Fig. 6.

As can be seen from Fig. 6, with the increase of the iterative times, the accuracy of the model is also constantly improving, and when the number of iterations reaches 20, the accuracy of the model gradually tends to be stable, and remains at about 0.983. The above results show that the MSCNN model has high accuracy, and can continuously improve the accuracy of the model with the increase of the number of iterations. Taking the mean value of 0 as the random initialization premise of all layers with very biased weights, the results of exploring the connection probability between adjacent layers are shown in Fig. 7.

Table 1
Error rate comparison of different pooling methods.

/	1	2	3	4	5	6	7	8	9	10
Avg	44%	32%	31%	24%	20%	16%	11%	7%	7%	5%
Max	51%	45%	36%	32%	27%	23%	17%	12%	9%	8%

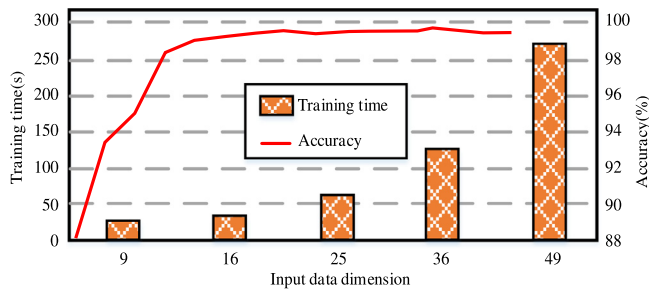


Fig. 8. Training time of different dimension models.

With the increasing of connection probability, the classification accuracy of MSCNN model is also increasing. When $P=0.7$, the accuracy rate reaches the maximum value of 95.7%. Then, with the increase of connection probability, the accuracy of MSCNN model does not increase and decrease. The results show that 0.7 should be chosen as the connection probability for the probability selection between the connection layers, which can greatly improve the accuracy. By analyzing the difference of pooling in the overlapping mode of the pooling layer, the step size is 1–10, and the comparison (Table 1).

As can be seen from Table 1, when average pooling is used to eliminate noise, the error rate decreases with the increase of step size, and decreases by 5% from the highest 44%. On the contrary, Max Polo method can also eliminate noise better, and the error rate decreases with the increase of step size. However, it is obvious that the error rate of Max Polo is higher than average Polo under the same step size. The final solution shows that the average pooling is correct, which improves the convergence speed of the model, and increases the accuracy of the model. Fig. 7 shows the training time change of the system under the change of input data dimension.

The training time of the system model changes with the dimension of the input data, and there is a positive correlation between the two as shown in Fig. 8. And it is not difficult to find that with the increasing dimension of input data, the growth trend of training time is increasing, until the dimension increases to more than 12, the accuracy gradually tends to be stable, and gradually keeps the accuracy within the maximum range. This means that the accuracy of the model is affected by the dimension, and with the increase of the dimension, the accuracy of the model also continues to increase, indicating that the performance of the model is also continuously improving.

In addition, the effect of establishing experimental analysis BN layer is shown in Fig. 9.

The MSCNN model without BN layer increases with the increase of step number, and the accuracy rate increases gradually (Fig. 9). When the number of steps is less than 1000, the accuracy rate rises rapidly, and then gradually slows down. When the steps reach 4000, it will become stable and the accuracy rate will be stable at about 98%. However, the accuracy of MSCNN model with BN layer increases with the increase of steps. However, unlike the model without BN, when the number of steps is less than 500, the accuracy of the model increases rapidly and the speed is obviously higher than that without BN model. When the number of steps reaches 2000, the accuracy rate has begun to stabilize,

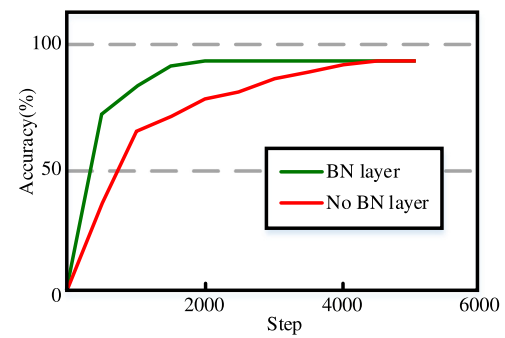


Fig. 9. Trend of detection accuracy.

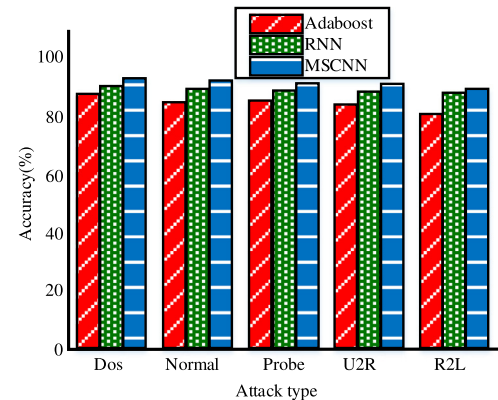


Fig. 10. Accuracy comparison of MSCNN to baseline methods.

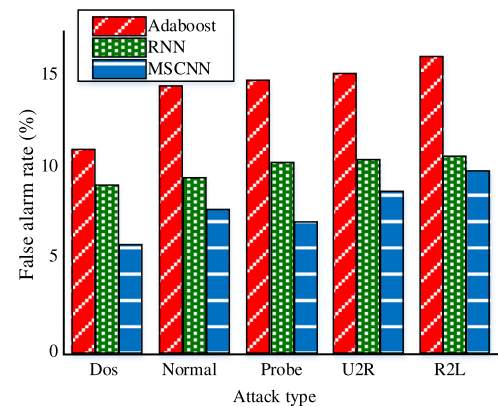


Fig. 11. Comparison of detection rates of three algorithms for five attack types.

keeping at about 98%. The results show that adding BN layer to MSCNN model effectively improves the convergence speed of the model and the learning speed of the model.

In addition, in order to verify the proposed model, it is compared to some baseline methods including Recurrent Neural Network (RNN) and Adaboost models as shown in Fig. 10.

The detection rate in Fig. 10 is the average detection rate of the three algorithms for different attack types. It can be seen from Fig. 10 that the three algorithms can well detect different attack types, and the detection rate is above 80%. The detection rates of Adaboost to DOS, Normal, Probe, U2R and R2L are 87.3%, 83.1%, 84.1%, 83.7% and 80.2% respectively. The detection rates of RNN to DOS, Normal, Probe, U2R and R2L are 90.3%, 90.3%, 90.2%, 90.3% and 90.4% respectively. By observing the detection rate of MSCNN model, it is found that the detection rate of MSCNN model for five

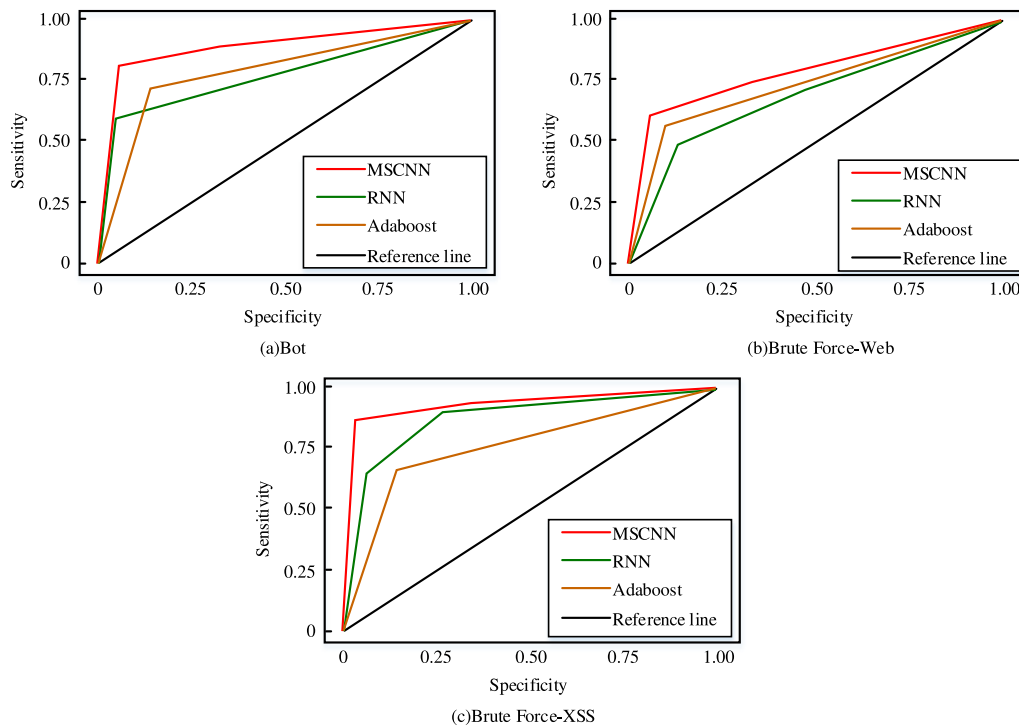


Fig. 12. Comparison of ROC curves for three algorithms against five attack types.

attack types is 94.2%, 93.4%, 92.5%, 92.1% and 90.3% respectively, which is 4.37% higher than that of other models. The results show that the detection rate of MSCNN model in the detection of five types of attacks is higher than 90%, and it is obviously better than the detection rate of the other two models in the five types of attacks, which indicates that the detection effect of the proposed MSCNN model is excellent. In addition, the false alarm rates of the three algorithms in intrusion detection of DOS, Normal, Probe, U2R and R2L are compared, as shown in Fig. 11.

The traditional RNN model and Adaboost [27,28] model are used to compare the performance with MSCNN model. The three algorithms use the same test set and experimental environment. The comparison results of the three algorithms in detection rate are shown in Fig. 11.

In Fig. 11, the false alarm rate of Adaboost model is significantly higher, the false alarm rate of DOS is 11%, and the false alarm rate of R2L has reached 16.7%, and the false alarm rate of other three attack types has reached 14%, 14.5% and 15.2%. The false alarm rate of RNN model were 8.9%, 9.1%, 10.1%, 10.3% and 10.9% respectively. The false alarm rate of MSCNN model were 6.1%, 7.8%, 7.4%, 8.8% and 10.1% respectively, which are significantly lower than the other two models, with an average decrease of 4.02%. The results show that MSCNN model can detect the attack types in intrusion system well, and also show that the performance of MSCNN model is better than other models. In addition, the three detection models are applied to other data sets for detection and analysis, and the ROC curves of the three models are drawn to solve the offline area, so as to analyze the universality of the prediction efficiency of the three models. The ROC curve is shown in Fig. 12.

As can be seen from Fig. 12, in three different data sets, the three models can make good prediction, and the offline area is far greater than 0.5, which indicates that the prediction efficiency of the three models is good, but it is not difficult to see that the offline area of MSCNN model is always greater than the other two models. The above results show that the MSCNN model has a strong universality, can be trained well in different data sets, and the prediction performance meets the ideal requirements.

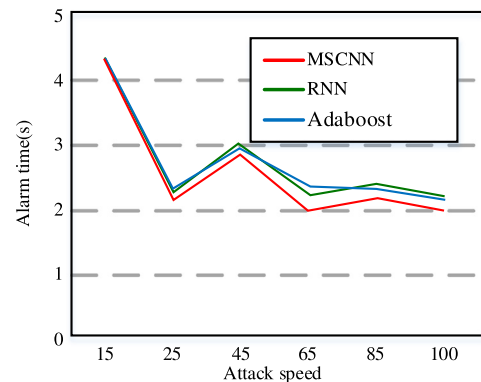


Fig. 13. Comparison of warning time under different detection methods.

For that, the short-term detection effect and long-term detection effect are analyzed respectively. First, the short-term application effect evaluation is carried out, and the average early warning time of intrusion detection based on different algorithms is compared, as shown in Fig. 13.

The warning time of the proposed detection system decreases with the increase of attack speed, which indicates that the intrusion detection system based on MSCNN can perfectly cope with external network attacks as shown in Fig. 13. The same experiment is also conducted for other two baseline methods, and the two detection models can also achieve reducing the alarm time with the increase of attack speed. However, from the overall situation, the early warning time of MSCNN intrusion detection system and the other two groups of detection systems is not different when the attack speed is low. However, as the attack speed increases by more than 25, it is obvious that the average early alarm time of MSCNN intrusion detection system is 0.1s and 0.15s faster than RNN and Adaboost respectively. The results show that MSCNN intrusion detection system after a short-term

Table 2
Multi-scale convolutional neural network intrusion detection system application effect.

IP number	Network unknown encrypted data transmission	Web service remote SQL injection attack	SOCKS proxy access operation	HTTP protocol URL field super long buffer overflow attack	Port scanner PING Sweep operation
1	17106	10321	5214	1321	996
2	20061	14679	9432	1782	1023
3	9752	7649	4125	922	813
4	15446	9983	4698	831	842
5	12043	9461	6894	1025	974
6	7342	5435	3283	1146	768
7	19413	11842	6749	1953	1102
8	11497	9238	5122	1846	953
9	21513	15230	9463	2145	1622
10	8846	7791	4131	1462	975

test, its response to network intrusion is faster, early alarm is fast, and with the speed of network attacks, it can also improve its own early warning time adaptively, which is conducive to network security protection.

And in practical application, the installation cost of the three intrusion detection systems and the later maintenance cost of MSCNN are slightly lower than the other two intrusion detection systems to a certain extent, so from the overall benefit, MSCNN has better usability. 10 IP addresses were selected as the test objects to be attacked. The number of network attacks and detection system in these 10 IP addresses in the next 6 months is shown in Table 2.

It can be seen from Table 2 that the 10 IP addresses are mainly affected by the network unknown encrypted data transmission, Web service remote SQL injection attack, SOCKS proxy access operation, HTTP protocol URL field super long buffer overflow attack and port scanner PING Sweep operates five types of attacks, in which the network unknown encrypted data transmission is the most frequent of all attacks, and the maximum number of attacks is 21513. The port scanner Ping sweep is the least attack type. The most attack is 1622 and the lowest is 768. It shows that under the support of intrusion detection system based on multi-scale convolutional neural network, the number of attacks of each IP is well detected. And it further shows that the intrusion detection system can achieve good results in practical application, and also verifies the effectiveness and feasibility of the research content.

5. Conclusion

In view of the defects of the network intrusion detection system in the market, a high-precision intrusion detection technology based on MSCNN is proposed. The performance of the network model and the effect of the detection system in the practical application are analyzed and discussed. MSCNN is a kind of deep learning method to realize high-precision recognition, and it is applied to the network security intrusion detection system. After the final performance analysis, it is found that the convergence rate of the network adding BN layer is significantly faster than that without BN layer. By comparing the accuracy and false alarm rate of different algorithms in the face of different attack types, it is found that the detection rate of MSCNN model is higher than that of other models when facing five attack types. In addition, the false alarm rate of MSCNN model detection is also obvious. In the practical application, it can be seen from the short-term application effect that the early alarm time of the system to network attack is better than other detection systems with the increase of attack speed. The long-term evaluation of application effect shows that the system can prevent and classify different types of attacks. In conclusion, MSCNN can effectively improve the intrusion detection, which provides a theoretical support for network security protection. However, in the future, there are still some rooms to improve the model to construct a better intrusion detection system.

CRediT authorship contribution statement

Jing Yu: Organize graphics, Modify articles, Reply to comments. **XiaoJun Ye:** Revise articles, Respond to comments, Submit Revised drafts. **Hongbo Li:** Organize references, Revise articles and reply to comments.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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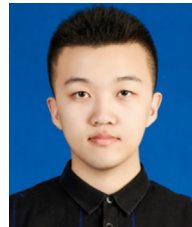
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